

Occurrence and Seasonal dynamics of the digenean, *Tremiorchis ranarum* Mehra et Negi, 1926 (Digenea: Plagiorchiidae) in the Indian bull frog, *Hoplobatrachus tigerinus* Daudin, 1803 from Kadapa District, Andhra Pradesh, India

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ABSTRACT

The Indian bull frog, *Hoplobatrachus tigerinus* Daudin, 1803 is frequently found to be infected with the digenean, *Tremiorchis ranarum* Mehra et Negi, 1926. The seasonal dynamics of the *T.ranarum* was studied in the frog during February, 2013 to February, 2014 from Kadapa District, Andhra Pradesh. Of the total of 130 frogs examined, only 18 frogs (13.84%) were infected with this digenean (n=530). Intensity of infection ranged from 1 to 229, with mean intensity (29.44 ± 20.82), mean abundance (4.07 ± 2.88) and index of infection (0.56). Monthly population dynamics of *T. ranarum* of *H. tigerinus* was analyzed in term of prevalence, mean intensity and mean abundance. Medium sized and larger frogs showed more infection and there is effect of sex on the parasitic infection of *T. ranarum*. Males are more infected when compared to females.

Keywords: *Hoplobatrachus tigerinus*, *Tremiorchis ranarum*, prevalence, mean intensity and meanabundance.

1. INTRODUCTION

Wild amphibian populations and their natural habitats are at stake due to global changes and severe pollution throughout the world which may also destroy their parasite biodiversity or may amplify the effect of pathogen and their diseases (Taylor et al., 2005; McKenzie, 2007; Tinsley et al., 2012). There are several areas in India, in particular in Southern India especially Kadapa where the biodiversity of metazoan parasites in vertebrate hosts is poorly known. Amphibians are one of the vertebrate hosts affected by the human activities. Amphibian hosts serve as the most preferred vertebrate hosts for a number of metazoan parasites like digeneans, monogeneans, cestodes, nematodes, acanthocephalans and protozoans. *Hoplobatrachus tigerinus* of the family Dicroglossidae Anderson, 1871 is the commonly occurring aquatic species of anurans inhabiting the natural and artificial (paddy fields) wetland areas of India (Alexander, 2011). *Tremiorchis ranarum* Mehra et Negi, 1925 is one of the most prevalent digenetic parasite infecting the Indian Bull frog, *Hoplobatrachus tigerinus*. In the present investigation, plagiiorchiid digenean *T. ranarum* was collected from the intestine of *H. tigerinus* from various places of Kadapa district of Andhra Pradesh to analyze their seasonal dynamics. An appreciable amount of work has been contributed worldwide on metazoan parasites of Amphibians (Mehra et Negi, 1926, 1928; Bhalerao, 1926; Verma, 1930; Srivastava, 1933; Fishchthal and Thomas, 1968; Fotedar, 1959a, 1959b; Agarwal, 1966, 1968; Jahan, 1973; Khan and Mohiuddin, 1968; Pandey and Chakravarti 1968; Mukerjee and Gosh 1970a, b, 1972; Pandey, 1973; Rao and Kameshwari, 1976; Simha, 1977; Hafeezullah and Dutta 1981, 1985; Diengdoh 1989; Diengdoh and Tandon, 1991; Prudhoe and Bray 1982; Verma and Singh, 2000; Luque et al., 2005; Lemke et al., 2008; Schotthofer et al., 2009; Mohammad et al., 2010). However, only a negligible amount of work has been carried out on the seasonal dynamics, morphological intra-specific variations and life-cycles of *T. ranarum* from India (Mehra et Negi, 1926, 1928; Bhalerao, 1926; Verma, 1930; Bhardwaj, 1962; Ali and Karyakarte, 1970; Dwivedi and Chauhan, 1970; Sinha and Sahay, 1971, Pandey, 1973; Kalyanker and Palladwar, 1978; Karyakarte and Baheti, 1980; Pandey and Agarwal, 1981; Madhavi et al., 1987; Ratnakumari; 2006 a, 2006 b; Hafeezullah and Dutta 1981; Rajendran and Janardanan, 1993; Sah and Sahay, 1998; Pandey et al., 2013). Insufficient amount of work has been contributed from Southern peninsular India and these sites require an additional revision or new comprehensive approach, in particular Kadapa region of Andhra Pradesh. Hence, a serious attempt was made to study the occurrence and seasonal dynamics in *T. ranarum* in *H. batrachus*.

2. MATERIALS AND METHODS

Sampling Sites

Kadapa District (Lat. 14°28'N 78°49'E, 137 m Altitude), located in Andhra Pradesh state was sampled during February, 2013 to February, 2014. The fieldwork was based in an Industrial estate areas, Ramapuram village and University campus areas where natural vegetations are disturbed by anthropogenic activities. *Hoplobatrachus tigerinus* (n=130), procured from these areas were transported alive to the laboratory and were anaesthetized before dissecting to examine the parasites under the stereozoom microscope (LM-52-3621 Elegant) from February, 2013 to February, 2014. Snout-vent length and sex of each frog was note down carefully and all organs were examined discretely for collection and counting. *H. batrachus* measured 4-18.5 cm (mean= 10.97±3.78) in total length. The average total snout-vent length of female (11.15±3.84 cm, n=46) and male (10.88±3.77 cm, n=84) frog in the sample were not significantly different (t= 0.69). Pearson's coefficient of correlation 'r' was applied to study the relationship between host's snout-vent length and parasitisation. The influence of host sex on the abundance and prevalence of parasites was analysed by applying Chi-square test. During the present study, only 530 parasites were collected which were cleansed thoroughly by ordinary tap water, killed and fixed quickly under pressure of the cover glass by using a fixative, A.F.A (85:10:5) for 24 hours and were washed thoroughly in running tap water for few minutes and stained with alum caramine. The specimens after being stained with Alum caramine were briefly washed in tap water and dehydrated through series of graded alcohols (70%, 90%, 95% and 100%) cleared in xylol and finally mounted in Canada balsam or DPX Mountant and due care is taken to avoid the formation of air bubbles during and after mounting (Hiware et al., 2003; Madhavi et al., 2007). Specific characterizations were carefully observed under the Lynx Trinocular microscope (N-800M) and figures are drawn with the aid of attached drawing tube. Conventional terms are used to describe the species. All measurements will be taken with the aid of an ocular micrometer in millimeters. Quantitative and qualitative analysis of the data were carried out by following the standard statistical books (Snedecor and Cochran, 1967; Sundara Rao and Richards, 1996; Daniel, 1998) and the statistical computations like prevalence, mean intensity, mean abundance, standard deviation were carried out using Microsoft Excel (Office 2007). Ecological terminology is as suggested by Margolis et al., 1982 and Bush et al., 1997.

3. RESULTS AND DISCUSSION

Taxonomic summary

Host: *Hoplobatrachus tigerinus* Daudin, 1926

Infection site: Intestine

Collection site: Kadapa

Body 2.2-4.10 × 0.47-1.07, small backwardly directed spines extend from anterior end to little posterior to hind end of the posterior testis. Cuticle thick and smooth. Oral sucker 0.12- 0.35 × 0.14-0.35. Prepharynx small, thin walled; pharynx 0.1-0.15 × 0.1-0.15. Fore body, 1.1-2.5 × 0.7-1.20; hind body, 1.1-2.5 × 0.55-0.95. Intestinal caeca simple extending up to anterior or hind end of anterior testis. Acetabulum, 0.15-0.25 × 0.14-0.22. Testis two, anterior testis, 0.3-0.37 × 0.2-0.42; posterior testis, 0.3-0.32 × 0.2-0.45. Cirrus sac, 0.54-0.55 × 0.14-0.15. Ovary, 0.20-0.28

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× 0.16-0.26. Vitelline follicles, 0.05-0.11× 0.025-0.10. Uterus small on posterior region. Eggs small and non operculated, 0.0125-0.0275 × 0.010 – 0.015.

Seasonal dynamics

Hoplobatrachus tigerinus is the extremely aquatic species of anurans inhabiting the natural and artificial (paddy fields) wetland areas of India (Alexander, 2011). The aquatic environment of frog permits the metacercariae larvae of plagiorchiid trematodes to enter their body when they feed on metacercariae infected tadpoles, resulting in infection (Rajendran and Janardhanan, 1993). Of the 130 hosts examined, only 18 frogs (13.84%) were found to be infected with 530 (range = 1-229) *T. ranarum*. It is a recurrently occurring plagiorchiid digenean infecting the intestines of the hosts. *T. ranarum* represented a prevalence of 13.84%, with mean intensity (29.44±20.82), mean abundance (4.07±2.88) and index of infection (0.56) (Table-1). Monthly changes in the prevalence, mean intensity, mean abundance and index of infection of *T. ranarum* were depicted graphically (Fig-1). Prevalence was high in the months of May, September, December to February, 2014 and almost nil in the months of July, October and November. Mean intensity, mean abundance and index of infection were high in the month of December as the maximum number of *T. ranarum* (429) was recovered from the 20 frogs examined. Prevalence of parasitization, mean intensity, abundance and index of infection was relatively higher in winter followed by rainy and summer (Fig.2). High population levels of *T. ranarum* in winter can be attributed to high levels of occurrence of hosts in winter months. Fall in temperature might be one of the causes for reduced immunological response in hosts. Coefficient of Correlation 'r' was designated to study the possible relationship between host size and parasitization. The computed value 'r' 0.019 illustrated a very insignificant positive correlation between host size and *T. ranarum* infection. Parasitization was relatively high in medium sized frogs (Group-II) and slightly less in smaller (Group-I) and larger (Group-III) frogs (Table-2). The present study supports the views of Aho (1990), Sluys et al., (1994), Koyun (2012) and Comas et al., (2014) that there is a positive relationship between body size and parasite intensity. Impact of host sex on the parasitic loads of amphibians is unpredictable among different host and parasite species and is seems to be more complex cause than the size of the host (Hollis, 1972; Kehr et al., 2000; Verma and Singh, 2000 and Sluys et al., 2006, Begum and Bhanu, 2012). Of the total sample of 130 *H. batrachus* screened, 84 (64.6%) were males and only 46 (35.3%) were females. Only 12 (14%) males and 06 (13%) were infected by these parasites (Table-3). Though females showed marginally high prevalence than males, however, there is significant sex-related difference in the parasitic abundance of *T. ranarum* between male and female *H. batrachus* which is evidenced by the calculated χ^2 -value (46.37) at 5% level of significance and 1° degree of freedom being greater than the tabulated χ^2 -value (3.84).

Table 1

Diversity parameters and distribution pattern of *Tremiorchis ranarum* in *H. batrachus*

Name of the parasite	Infected frogs	Total no. of parasites	Prevalence (%)	Mean Intensity	Mean abundance	Index of infection	Range	Location
<i>Tremiorchis ranarum</i>	18	530	13.84	29.44±20.8	4.07±2.88	0.56	1-229	Intestine

Table 2

Correlation coefficient (r) between size and parasitic number of *T. ranarum* in *H. batrachus*

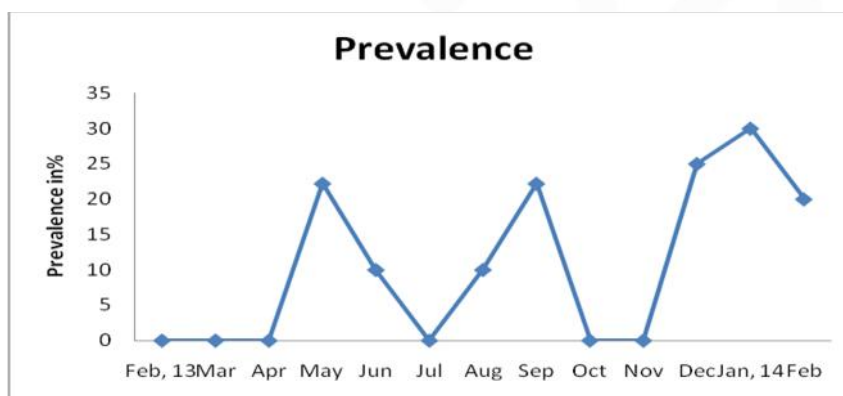
Sl.No.	Size groups	Class intervals	No. of parasites	Correlation coefficient (r)
1	Group-I	4-10	144	r = 0.019
2	Group-II	10-16	241	
3	Group-III	16-22	145	

Table 3

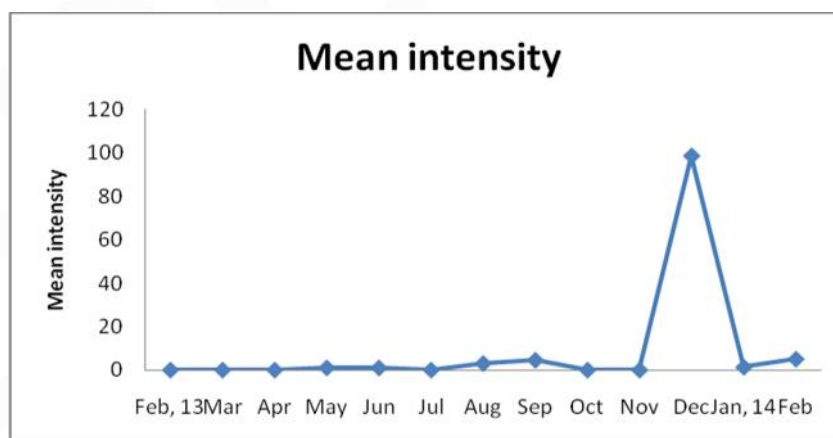
Parasitization of *T. ranarum* in males and females of *H. batrachus* ($N_m=84$; $N_f=46$)

Name of the species	N_{mi}	N_{fi}	P_m	P_f	M_{im}	M_{if}	MA_m	MA_f
<i>T. ranarum</i>	12	6	14.28	13.04	43	2.33	6.14	0.30

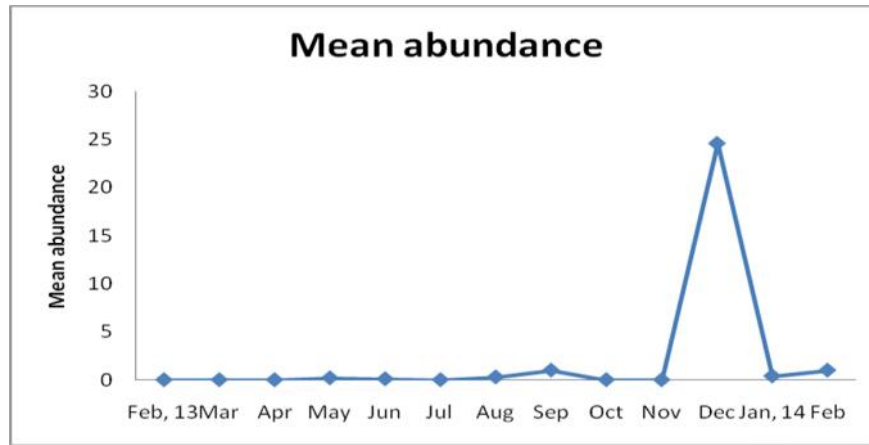
N_m number of males examined, N_f number of females examined, N_{mi} number of infected males, N_{fi} number of infected females, P_m and P_f prevalence of males and females respectively, M_{im} & M_{if} mean intensity of males and females; MA_m & MA_f mean abundance of males and females.



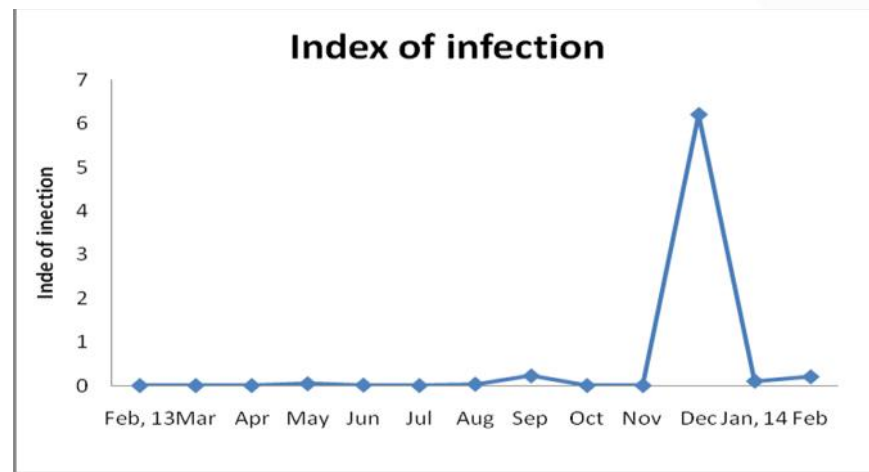
a.



b.



c.



d.

Figure 1

Monthly population changes of *Tremiorchis ranarum* in *H. batrachus*; **a.** Prevalence, **b.** Mean intensity, **c.** Mean abundance, **d.** Index of infection

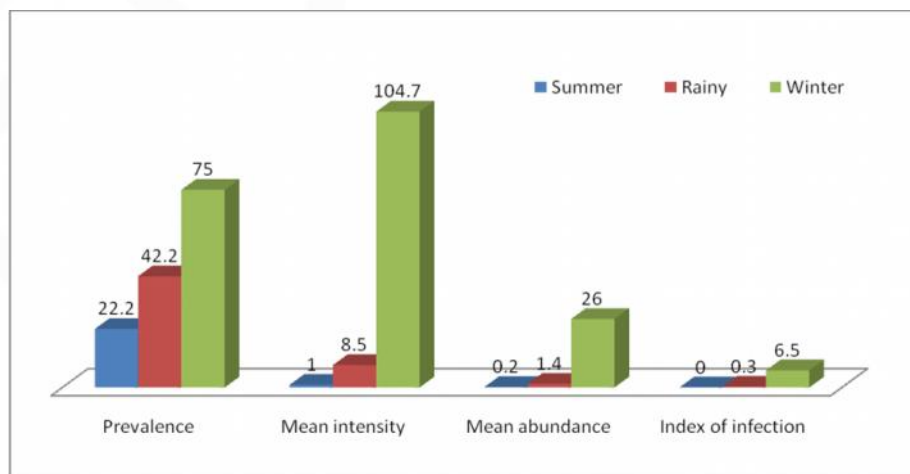


Figure 2

Seasonal variations in prevalence, mean intensity, mean abundance & index of infection of *T. ranarum* in *H. batrachus*

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REFERENCES & RESOURCES

1. Agarwal V. Studies on some trematodes of frogs from Lucknow. *Indian Journal of Helminthology*, 1966, 28 (1), 82-90
2. Agarwal V. On some trematode parasites of amphibia from Lucknow. *Rev Biol Trop*, 1968, 15(1), 1-11
3. Aho JM. Helminth communities of amphibians and reptiles: comparative approaches to understanding patterns and processes. In: GW ESCH, A.O.BUSH & J.M.AHO (eds.), *parasite communities: patterns and processes*. Chapman and Hall, New York. 1990, 157-195
4. Alexander R. Indian bull frog, *Hoplobatrachus tigerinus* (Daudin, 1803) of Ousteri Lake, Puducherry. *Current Science*, 2011, 100 (6), 806
5. Ali SM, Karyakarte PP. Observations on developmental stages of *Tremiorchis ranarum* Mehra et Negi, 1926 (Trematoda: Plagiorchiidae). *Marathwada University Journal sci*, 1970, 9, 161-168
6. Begum A, Banu N. Sex, organal and seasonal differences of helminthofauna of Toad, *bufo melanostictus* (Schneider, 1799). *Bangladesh J Zool*, 2012, 40 (2), 155-164
7. Bhalerao GD. On the Trematodes of the Digestive Tract of a common Indian Frog, *Rana tigrina*, with a description of *Centrovitus pentadelphi* n.g., n.sp. *Parasitology*, 1926, 18 (02)
8. Bhardwaj ON. Studies on the intraspecific variations of *Tremiorchis ranarum* Mehra et Negi, 1926, Plagiorchiidae Luhe, emert Ward, 1917, Trematoda and discussion on causes and effects of intraspecific variations. *Proc Nat Acad Sci*, 1962, 32, 445-450
9. Bush AO, Lafferty KD, Lotz JM, Shostak AW. Parasitology meets ecology in its own terms: Margolis et al. 1962. *Revisited J Parasitol*, 1997, 83(4), 575-583
10. Comas M, Ribas A, Mialzo C, Sperone E, Triepi S. High levels of prevalence related to age and body condition: Host-parasite interaction in a water frog, *Pelophylax kl. hispanicus*. *Acta herpatologica*, 2014, 9 (1), 25-31
11. Daniel WW. Biostatistics: A foundation for analysis in the Health Sciences, 1998, 8th edn. Wiley, Newyork
12. Diengdoh CR, Tandon V. A new species of *Polystoma* (Monogenea) parasites in rhacophorid amphibians in Meghalaya, India. *Helminthologia*, 1991, 28, 173-178
13. Diengdoh CR. Helminth parasite spectrum of Amphibian Hosts in Meghalaya. Ph.D. thesis. North- Eastern Hill University (N.E.H.U) Shillong, 1989, 129
14. Dwivedi MP, Chauhan BS. On some digenetic trematodes. Part-II. *J Zool Soc India*, 1970, 22, 87-96
15. Fishthal JH, Thomas JD. Digenetic trematodes of amphibians and reptiles from Ghana. *Proc Helm soc Wash*, 1968, 35, 1-15
16. Fotedar DN. On a new species of the genus *Ganeo* Klein, 1905 and some notes on the genus. *J Helminthol*, 1959a, 33, 151-160
17. Fotedar DN. On a new species of the genus *Cosmocerca* Diesing, 1861 from a toad, *Bufo viridis*, in Kashmir. *J Helminthol*, 1959b, 33, 161-170
18. Hafeexullah M, Dutta IB. Studies on some trematodes (Digenea: Lecithodendriidae) of Pisces and amphibian from east coast of India. *Bull Zool sur Ind*, 1985, 4(2), 69-82
19. Hafeexullah M, Dutta IB. On the Monotype of the genera *Anchitrema* Looss, 1899 and *Tremiorchis* Mehra and Negi, 1926 with a note on *Pleurogenoides-Indopleurogenes- Allopleurogenes* complex. *Bull Zool sur Ind*, 1981, 4(2), 165-171
20. Hiware CJ, Jadhav BV, Mohekar AD. Applied Parasitology. A practical manual. Mangaldeep, Jaipur. 2003.
21. Hollis PD. Host sex influence on seasonal incidence of *Haematoloechus mediplexus* (Trematoda: Plagiorchiidae) in *Rana pipens*. *J Parasitol*, 1972, 58(1), 128
22. Jahan A. Studies on digenetic trematodes of vertebrates. Part II. A trematode, *Gorgoderina guptai* n.sp. from the urinary bladder of a toad from Lucknow. *Japanese Journal of parasitology*, 1973, 22, 259-261
23. Kalyanker SD, Palladwar VD. Notes on the taxonomic status of the genus *Tremiorchis ranarum* Mehra et Negi, with a description of the new species of the genus from Indian frog. *Marath Univ J Sci*, 1978, 16, 229-232
24. Karyakarte PP, Baheti SP. Studies on free aminoacids in *Tremiorchis ranarum*, Mehra and Negi, 1926 (Trematoda: Digenea). *Rivista di parasitologia*, 1980, 41(1), 81-83
25. Kehr M, Manly BFJ, Hamann MI. Coexistence of helminth species in *Lysapsus limellus* (Anura: Pseudidae) from an Argentinian subtropical area: influence of biotic and abiotic factors. *Oecologia*, 2000, 125, (4), 549-558
26. Khan MM, Mohiuddin A. Notes of parasites of frogs, *Rana cyanophlyctis* Schneider and *Rana tigrina* Daud. Proceedings of 20th Pakistan Science congress. Abstracts (3) Biology C 1968, 35-36
27. Koyun M. The occurrence of parasitic helminths of *Capoeta umbla* in relation to seasons, host size, age and gender of the host in Murat River. *Turkey J Anim. Vet Adv*, 2012, 11, 609-614
28. Lemke LB, Dronen, N, Fox JG, Prashant RN. Infestation of Wild-caught American Bullfrogs, *Rana catesbeiana* by multiple species of Metazoan parasites. *J American Ass for Lab Anim Sci*, 2008, 4, 42-46.
29. Luque JL, Martins AN, Tavares LER. Community structure of metazoan parasites of the yellow Cururu toad, *Bufo ictericus* (Anura: Bufonidae) from Rio de Janeiro, Brazil. *Acta Parasitol*, 2005, 50 (3), 215-220
30. Madhavi R, Dhanakumari C, Ratnakumari TB. The life history of *Pleurogenoides orientalis* (Srivastava, 1934) (Trematoda: Lecithodendriidae). *Parasitol Res*, 1987, 73, 41-45
31. Madhavi R, Vijayalakshmi C, Shyamasundari K. Collection, staining and identification of different helminth parasites. A manual of the workshop on fish parasites – taxonomy capacity building. Andhra University Press, Visakhapatnam. 2007.
32. Margolis L, Esch GW, Holmes JC, Kuris AM, Schad GA. The use of ecological terms in parasitology (report of an Adhoc use of committee of the American Society of Parasitologists). *J Parasitol*, 1982, 68, 131-137
33. McKenzie VJ. Human land use and patterns of parasitism in tropical amphibian hosts. *Biol Con*, 2007, 137, 102-116
34. Mehra HR, Negi PS. On a new trematode *Tremiorchis ranarum* nov.gen. nov. sp. from common Indian frog *Rana tigrina*. *Parasitology*, 1926, 18, 168-181
35. Mehra HR, Negi PS. Trematode parasites of the Pleurogenetinae from *Rana tigrina*, with a revision and synopsis of the subfamily. *Allah Univ Studies*, 1928, 4, 63-118
36. Mohammad KM, Al-Moussawi AA, Jasim SY. Helminth Parasites of the Green Toad *Bufo viridis* Laurenti, 1768 in Baghdad Area, Central Iraq. *Egypt Acad J biolog Sci*, 2010, 2(1), 17-25
37. Mukherjee RP, Ghosh RK. Studies on some amphibian trematodes from West Bengal and Maharashtra (part II). *Rec Zool survey Ind*, 1972, 66, 273-276
38. Mukherjee RP, Ghosh RK. A note on some abnormal characters of *Pleurogenoides gastroporus* (Trematoda: Lecithodendriidae). *Sci & cult*, 1970b, 36, 561-562
39. Mukherjee RP, Ghosh RK. Studies on some amphibian trematodes from Uttar Pradesh and West Bengal (Part I). *Ind J Helminthol*, 1970a, 2(1), 61-78
40. Pandey D, Pandey JP, Shukla N. Morphological intraspecific variations of *Tremiorchis ranarum* of *Rana tigrina* in District Sidhi (M.P) India. *Int J Scient. Res Public*, 2013, 3(5), 1-3
41. Pandey KC, Agarwal N. Further remarks on the genus *Tremiorchis* Mehra et Negi, 1925. *Proc Ind Acad sci*, 1981, 90(5), 567-570
42. Pandey KC, Chakrabarti KK. On a new trematode from the common Indian frog, *Ganeo gazipurensis* n. sp. *Sci & Cult*, 1968, 34, 218-219
43. Pandey KC. Studies on some known trematodes parasites. *Ind J Zool*, 1973, 14, 192-219

44. Prudhoe S, Bray RA. Platyhelminth parasites of the Amphibia. Oxford: *British Museum (Natural History); Oxford University Press*, 1982
45. Rajendran KV, Janardanan KP. Studies on the life-cycle of *Tremiorchis ranarum*. *Journal of Helminthology*, 1993, 67(2), 95-101
46. Rao LN, Kameshwari M. On the synonymy of *Ganeo korkei* Bhalerao, 1936 and *Ganeo panjabensis* Gupta, 1954 with *Ganeo tigrinum* Mehra and Negi, 1928. *Current science*, 1976, 45(17), 623-628
47. Ratnakumari TB. Population biology of the plagiorchiid trematode, *Tremiorchis ranarum* (Mehra), Negi 1926) in the Skipper frog *Rana cyanophlyctis* Schneider. *Bull pure appl sci-sect (A)*, 2006a, 25 (2), 113-22
48. Ratnakumari TB, Intraspecific variations in *Tremiorchis ranarum* Mehra and Negi, 1926, Trematodes: Plagiorchidae). Schneider. *Bull Pure Appl Sci. (A)*, 2006b, 26 (1), 15-22
49. Sah HCP, Sahay U. A discussion on the synonymy of *Tremiorchis ranarum* Mehra Rai 1962; *T. vitelloconfluentum* Rai, 1962 with *Tremiorchis ranarum* Mehra et Negi, 1926. 10th All India congress of zoology, 14th -18th October, 1998, 40-46
50. Schothoefer AM, Bolek MG, Cole RA, Beasley VR. Parasites of the Mink frog (*Rana septentrionalis*) from Minnesota, U.S.A. *Comp parasitol*, 2009, 76(2), 240-246
51. Simha M. Studies on the amphibian trematodes from Tamilnadu and Kerala. *Rec Zool survey Ind*, 1977, 72, 291-294
52. Sinha A, Sahay U. On the occurrence of *Tremiorchis ranarum* Mehra et Negi, 1926 from *Rana cyanophlyctis* at Putna. *Ind J Helminthol*, 1971, 23, 56-69
53. Sluys MV, Rocha CFD, Ribas SC. Nematodes infecting the Lizard *Tropidurus itambere* in southeastern Brazil. *Amphibia-Reptilia*, 1994, 15(4), 405-408
54. Sluys MV, Schittini GM, Marra RV, Azevedo ARM, Vincente JJ, Vrcibradic D. Body size, diet and endoparasites of the Microhylid frog, *Chiasmocleius capixaba* in an Atlantic forest area of Southern Bahia state, Brazil. *Braz Biol*, 2006, 66 (1A), 167-173
55. Snedecor WG, Cochran GW. Biostatistical methods, 6th edn. Iowa state University press, Iowa, 1967, P593.
56. Srivastava HD. On a new trematodes of frogs and fishes of the United Provinces, India. Part I. Distomes of the family Hemiuridae from North India fishes and frogs with asystematic discussion on the family Halipegidae and the genera *Vitellotrema* Guberlet and *Genarchopsis* Ozaki. *Bull Acad science UP Agra & Oudh*, 1933, 3(1), 41-60
57. Sundara Rao PSS, Richard F. An introduction to biostatistics. A manual for students in Health sciences. Prentice-Hall, New Delhi. 1996
58. Taylor B, Skelly D, Demarchis LK, Slade MD, Galusha D, Rabinowitz PM. Proximity to pollution source and risk of amphibian limb malformation. *Env Health Persp*, 2005, 113, 1497-1501
59. Tinsley RC, York JE, Stott LE, Everard AL, Chapple SJ, Tinsley MC. Environmental constraints influencing survival of an African parasite in a north temperate habitat: effects of temperature on development within host. *Parasitol*, 2011, 138, 1039-1052
60. Verma SC. On the synonymy of the Genera *Tremiorchis* Mehra and Negi 1926 and *Centrovitus* Bhalerao 1926, with the description of *Tremiorchis ranarum* n.sp. *Parasitol*, 1930, 22 (03)
61. Verma AK, Singh, G. A Quantitative analysis of Gastro-intestinal helminths (Trematode: Digenea) infection in ranid frogs of Jammu. *Zoos' print journal*, 2000, 15(4), 233-238